

## Quotient Space

Def. Let  $(X, \mathcal{C})$  be a Topological Space, and  $Y$  be a set.

For an onto map  $f: X \rightarrow Y$ , define a Quotient Topology on  $Y$  induced by  $f$  as

$$\mathcal{C}_f \stackrel{\text{def}}{=} \{U \subseteq Y \mid f^{-1}[U] \text{ is open of } (X, \mathcal{C})\}.$$

It is actually a Topology on  $Y$ , because: clearly,  $\emptyset, Y \in \mathcal{C}_f$  and

$$U_\alpha \in \mathcal{C}_f \ (\alpha \in I) \Rightarrow f^{-1}\left[\bigcup_{\alpha \in I} U_\alpha\right] = \bigcup_{\alpha \in I} f^{-1}[U_\alpha] \text{ is also open in } X, \text{ so is } \bigcup U_\alpha.$$

Similarly,  $f^{-1}\left[\bigcap_{\alpha \in I} U_\alpha\right] = \bigcap_{\alpha \in I} f^{-1}[U_\alpha]$  also open if  $I$  is finite, so is  $\bigcap U_\alpha$ .

Moreover, it is the finest Topology on  $Y$  such that  $f$  is Continuous, because

$$f: X \rightarrow Y \text{ is continuous iff } \forall U \in \mathcal{C}_Y, f^{-1}[U] \in \mathcal{C}_X \text{ iff } \mathcal{C}_Y \subseteq \mathcal{C}_f.$$

where  $\mathcal{C}_Y$  is arbitrary Topology on  $Y$ .  $\square$

Def. Let  $X$  be a Topological Space, and  $\sim$  be an equivalence relation on  $X$ .

Define the Canonical map on  $X$  as

$$\pi: X \rightarrow X/\sim : x \mapsto [x]$$

The  $(X/\sim, \mathcal{C}_\pi)$  is called Quotient Space where  $\mathcal{C}_\pi$  is a Quotient topology on  $X/\sim$ , induced by  $\pi$ . That is,

$$\mathcal{C}_\pi = \{U \subseteq X/\sim \mid \pi^{-1}[U] \text{ is open}\},$$

Note: For any  $U \subseteq X/\sim$ ,  $\pi^{-1}[U] = \{x \in X \mid \pi(x) \in U\} = \bigcup_{[x] \in U} [x]$

and for any  $A \subseteq X$ ,  $\pi^{-1}[\pi[A]] = \bigcup_{x \in A} [x]$ .

$\stackrel{\text{11 def}}{\text{Sat}} f(A)$ ,  $f$ -saturation of  $A$ .

Let  $X$  be a Topological Space,  $\sim$  be an equivalence relation on  $X$ ,  $\pi$  be a Canonical map.  
 Lemma. Let  $Z$  be a Topological Space and a map  $g: X/\sim \rightarrow Z$ ,

$g$  is Continuous if and only if  $g \circ \pi$  is Continuous.

Proof. Since  $\pi: X \rightarrow X/\sim$  is Continuous, trivially  $g \circ \pi$  is Continuous.

Conversely, assume  $g \circ \pi$  is Continuous. Then, given an open  $U \subseteq Z$ ,

$(g \circ \pi)^{-1}[U] = \pi^{-1}[g^{-1}[U]]$  is open of  $X$ , so  $g^{-1}[U]$  is open of  $X/\sim$ .  $\square$

Lemma. Let  $Z$  be a Topological Space.

If a Continuous map  $f: X \rightarrow Z$  satisfies  $x \sim y \Rightarrow f(x) = f(y)$ ,  
 then a map induced by  $f$ :

$$\bar{f}: X/\sim \rightarrow Z: [x] \mapsto f(x)$$

is Continuous and it is a Unique map such that  $\bar{f} \circ \pi = f$ .

Proof.  $\bar{f}$  is well-defined by:  $[x] = [y] \Leftrightarrow x \sim y \Rightarrow f(x) = f(y)$ .

Given  $x \in X$ ,  $f(x) = \bar{f}([x]) = \bar{f}(\pi(x)) = (\bar{f} \circ \pi)(x)$ , so  $\bar{f} \circ \pi = f$

By the Lemma above,  $\bar{f}$  is Continuous. And, given a map  $g: X/\sim \rightarrow Z$  s.t.  $g \circ \pi = f$ ,

$$g([x]) = g(\pi(x)) = f(x) = \bar{f}([x])$$

So the Uniqueness is guaranteed.

Lemma. Let  $Z$  be a Topological Space.

If a Continuous onto map  $f: X \rightarrow Z$  satisfies  $x \sim y \Leftrightarrow f(x) = f(y)$  and either open or closed  
 then a map induced by  $f$ :

$$\bar{f}: X/\sim \rightarrow Z: [x] \mapsto f(x)$$

is Homeomorphism.

Quotient map.

Def. Given Topological space  $X$  and  $Y$ , a map  $f: X \rightarrow Y$  is called a quotient map if:  
 $f$  is Continuous, Onto, and satisfies for any  $U \subseteq Y$ ,

$U$  is open in  $Y \xLeftrightarrow[\text{Quotient}]{f \text{ Continuous}} f^{-1}[U]$  is open in  $X$ .

Prop. Composition of Quotient map is also Quotient map.

Proof. Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be Quotient maps. Given open set  $U \subseteq Z$ ,

$(g \circ f)^{-1}[U]$  is open  $\Leftrightarrow f^{-1}[g^{-1}[U]]$  is open  $\Leftrightarrow g^{-1}[U]$  is open  $\Leftrightarrow U$  is open

And,  $(g \circ f)[X] = g[f[X]] = g[Y] = Z$ ,  $(g \circ f)$  is a Quotient map.  $\square$

Prop. If a Continuous Onto map  $f: X \rightarrow Y$  is either open or closed map, then  $f$  is Quotient map.

Proof. Given  $U \subseteq Y$ , if  $f^{-1}[U]$  is open then  $f[f^{-1}[U]] = U$  is open, if  $f$  is open map.

And, if  $f^{-1}[U]$  is open, then  $f[X \setminus f^{-1}[U]]$  is closed if  $f$  is closed map.

Since  $f$  is onto,  $X \setminus f^{-1}[U] = f^{-1}[Y \setminus f^{-1}[U]] = f^{-1}[Y \setminus U]$ , so

$f[X \setminus f^{-1}[U]] = f[f^{-1}[Y \setminus U]] = Y \setminus U$  is closed, or  $U$  is open.  $\square$

But, there is a quotient map neither open nor closed: define  $X = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ or } y = 0\}$ .

$f: X \rightarrow \mathbb{R}; (x, y) \mapsto x$

Then, it is clearly onto, and Continuous is given by: for any  $a < b$  in  $\mathbb{R}$ ,

$f^{-1}[(a, b)] = [(a, b) \times \mathbb{R}] \cap X$  is open.

To show  $f$  is quotient map, suppose that  $f^{-1}[U]$  is open for some  $U \subseteq \mathbb{R}$ .

Given  $x \in U$ ,  $f((x, 0)) = x \in U$ , so  $(x, 0) \in f^{-1}[U]$ . Since  $f^{-1}[U]$  is open, for some  $\epsilon > 0$ ,  $(x, 0) \in B_\epsilon((x, 0)) \cap X \subseteq f^{-1}[U]$ . Thus,  $f[B_\epsilon((x, 0))] = (x - \epsilon, x + \epsilon) \cap f[X] = (x - \epsilon, x + \epsilon) \subseteq U$ .  
So,  $U$  is open.

Moreover,  $B_1((0, 2)) \cap X$  is open of  $X$ , but  $f[B_1((0, 2)) \cap X] = [0, 1]$ , not open.

And,  $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$  is closed because  $F' = \emptyset$  so  $F = \overline{F}$ .

But,  $f[F] = \{\frac{1}{n} \mid n \in \mathbb{N}\}$  is not closed in  $\mathbb{R}$ , since  $0 \notin f[F]$  but  $0 \in f[F]'$ .

Comment: A projection  $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}$  cannot be Closed map:  $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$ .

Prop. Let  $X$  be a Topological space and  $\sim$  be an equivalent relation on  $X$ .  
If the Canonical Projection  $\pi: X \rightarrow X/\sim: x \mapsto [x]$  is open map,  
then for a basis  $\mathcal{B}$  for  $X$ ,  $\mathcal{B}_\sim = \{\pi[B] \mid B \in \mathcal{B}\}$  is a basis for  $X/\sim$ .

Proof. Let  $\mathcal{U} \subseteq X/\sim$  be an open set, and let  $[p] \in \mathcal{U}$  be given.

Since  $\mathcal{U}$  is open,  $\pi^{-1}[\mathcal{U}] \subseteq X$  is open in  $X$ .

Meanwhile, for  $p \in [p]$   $\pi(p) = [p] \in \mathcal{U}$ , so  $p \in \pi^{-1}[\mathcal{U}]$ . Thus, there is  $B \in \mathcal{B}$  such that

$$p \in B \subseteq \pi^{-1}[\mathcal{U}].$$

Now, since  $\pi$  is onto,

$$[p] = \pi(p) \in \pi[B] \subseteq \pi[\pi^{-1}[\mathcal{U}]] = \mathcal{U}$$

Since  $\pi$  is open map,  $\pi[B]$  is open, so  $\mathcal{B}_\sim$  is a basis for  $X/\sim$ . ~~QED~~

Note: so, if  $\pi$  is open, then  $X$  2nd-countable  $\Rightarrow X/\sim$  ".

Examples.

Define equivalence relation  $\sim$  on  $\mathbb{R}$  generated by  $x \sim y \Leftrightarrow x - y \in \mathbb{Z}$ .  
Then,  $\mathbb{R}/\sim = \mathbb{R}/\mathbb{Z} = \{a + \mathbb{Z} \mid a \in \mathbb{R}\} = \{a + \mathbb{Z} \mid a \in [0, 1)\}$ .

$U \subseteq \mathbb{R}/\sim$  is open set if and only if  $\pi^{-1}[U]$  is open,

where  $\pi: \mathbb{R} \rightarrow \mathbb{R}/\sim; x \mapsto x + \mathbb{Z}$ .

So, the topology  $\mathcal{O}_{\sim}$  on  $\mathbb{R}/\sim$  is generated by  $\mathcal{B}_{\sim} = \{(a, b) + \mathbb{Z} \mid a < b\}$ , because:

Given  $(a, b) \subseteq \mathbb{R}$ ,  $\pi[(a, b)] = (a, b) + \mathbb{Z} = \{r + \mathbb{Z} \mid a < r < b\}$ ,

and  $\text{Sat}_{\pi}(a, b) = \pi^{-1}[\pi[(a, b)]] = \bigcup_{r \in (a, b)} r + \mathbb{Z} = \bigcup_{n \in \mathbb{Z}} (a + n, b + n)$ .

Since the saturation is open, so  $\pi[(a, b)]$  is also open. Therefore, the Canonical Projection  $\pi$  is open map, so the  $\mathcal{B}_{\sim}$  is basis for  $\mathbb{R}/\sim$ .

Moreover,  $\mathbb{R}/\sim$  is Homeomorphic to  $S^1$ , because i define a map

$$f: \mathbb{R} \rightarrow S^1; x \mapsto e^{2\pi i x}.$$

Then,  $f$  is continuous onto and open map satisfying

$$x \sim y \Leftrightarrow x - y \in \mathbb{Z} \Leftrightarrow e^{2\pi i(x-y)} = 1 \Leftrightarrow f(x) = f(y)$$

So, the induced map

$$\tilde{f}: \mathbb{R}/\sim \rightarrow S^1; [x] \mapsto f(x)$$

is well-defined and injective, and since  $f$  and  $\pi$  are onto and  $\tilde{f} \circ \pi = f$ ,

$$\tilde{f}[\pi^{-1}[U]] = \tilde{f}[\pi[x]] = (\tilde{f} \circ \pi)[x] = f[x] = U.$$

So  $\tilde{f}$  is also surjection. And, given open  $U \subseteq S^1$ ,  $\tilde{f}^{-1}[U]$  is open and

$$\tilde{f}^{-1}[U] \text{ is open} \Leftrightarrow \pi^{-1}[\tilde{f}^{-1}[U]] \text{ is open} \Leftrightarrow f^{-1}[U] \text{ is open}$$

Therefore  $\tilde{f}$  is also continuous. Finally, since the  $f$  is open map, given open  $V \subseteq \mathbb{R}/\sim$ ,

$$\tilde{f}[V] = \tilde{f}[\pi[\pi^{-1}[V]]] = f[\pi^{-1}[V]] \text{ is open, so}$$

Continuous open bijective map  $\tilde{f}$  is Homeomorphism.  $\square$